Introduction to Cloud Computing Lesson

Skills you will learn

IRONHACK

Lesson Overview

Cloud computing lets organisations **rent IT capabilities on demand** instead of buying and running their own hardware. In this lesson you will explore the key ideas that underpin every major cloud provider.

Learning Objectives

By the end of this lesson you should be able to **describe, compare, and justify**:

1. The main business and technical drivers behind cloud adoption.
2. The three service models—**IaaS, PaaS, SaaS**—and who is responsible for what in each one.
3. The four deployment models—**Public, Private, Hybrid, Multi-Cloud**—and when each makes sense.
4. The **shared-responsibility model** (security and operations) that applies to every cloud service.

> ***Tip for instructors:*** *use the "Objectives ↔ Assessment" alignment—every objective is later tested in the Knowledge-Check section.*

1 What Is Cloud Computing?

> *"Someone else's computer—plus automation, massive scale, and an SLA."*

Cloud computing is the **on-demand delivery of IT resources via the internet with pay-as-you-go pricing**. Typical services include:

- **Compute:** virtual machines (VMs), containers, serverless functions
- **Storage:** object, block, and file storage; databases; data warehouses
- **Networking:** virtual networks, load balancers, CDN, VPN, direct links
- **Advanced services:** IoT, AI/ML, analytics, messaging, security tooling

These resources live in globally distributed **data-centres (regions & availability zones)** operated by providers such as AWS, Microsoft Azure, or Google Cloud Platform.

2 Service Models & the Shared-Responsibility Line

Layer	On-Prem	IaaS	PaaS	SaaS
Data	🟦 You	🟦 You	🟦 You	🟦 You
Applications	🟦 You	🟦 You	🟦 You	🟧 Provider
Runtime / Middleware	🟦 You	🟦 You	🟧 Provider	🟧 Provider
OS	🟦 You	🟦 You	🟧 Provider	🟧 Provider
Virtualisation	🟦 You	🟧 Provider	🟧 Provider	🟧 Provider
Servers, Storage, Network, DC	🟦 You	🟧 Provider	🟧 Provider	🟧 Provider

🟦 *You = customer's responsibility* | 🟧 *Provider = managed by provider*

Quick definitions

- **IaaS** – You rent raw building blocks (VMs, disks, networks). Ideal when you need maximum control or must lift-and-shift legacy workloads.
- **PaaS** – The provider supplies the runtime and auto-scales it; you bring code. Speeds up development and frees ops teams from patching OSs.
- **SaaS** – A finished application accessed via browser or API. No servers, no patches—just configure and use (e.g., Salesforce, Office 365).

3 Deployment Models

Model	Typical Use Cases	Key Trade-offs
Public Cloud	Start-ups, bursting workloads, global web apps	Fast time-to-market, shared infra, possible compliance hurdles
Private Cloud	Regulated industries, latency-sensitive systems	Full control & customisation, high CapEx/OpEx
Hybrid Cloud	Gradual migrations, DR, sensitive data + big-data analytics	Flexibility, but requires integration & unified governance
Multi-Cloud	Avoid vendor lock-in, best-of-breed services, geo-redundancy	Redundancy & choice, but adds complexity & cross-skill demands

4 Business Benefits (and Reality-Checks)

Benefit	Why It Matters	Watch Out For
Cost Optimisation	From CapEx to OpEx; scale down to £0	Runaway costs without tagging & budgets
Elastic Scalability	Auto-add capacity during peaks	Poorly tuned auto-scaling wastes money
Global Reach	Deploy near customers to reduce latency	Data-sovereignty laws vary by region
Rapid Innovation	New services launched weekly	Feature sprawl & skill-gaps
Security & Compliance	Provider invests billions in security	*You* must still configure IAM, patch apps

5 Hands-On Lab (Optional, 30 min)

> ***Goal:*** *spin up an IaaS VM, deploy a container with PaaS, and consume a SaaS service, all in one free-tier AWS account.*

1. **IaaS:** Launch an Amazon EC2 t3.micro instance running Ubuntu.
2. **PaaS:** Push a sample Node.js app to AWS Elastic Beanstalk.
3. **SaaS:** Store a file in Amazon S3 and share the pre-signed URL.
4. **Compare:** Where did you configure firewalls, patches, backups?

6 Further Reading & Practice

1. **AWS Well-Architected Framework – Cost and Reliability pillars**
2. **Microsoft's PaaS vs. IaaS whitepaper (June 2025 edition)**
3. **Google Cloud Skills Boost: Introduction to Cloud Foundations (free quest)**
4. **Coursera "Architecting Hybrid Cloud Infrastructure" mini-course**

8 Summary Cheat-Sheet (printable)

> ***Cloud computing*** *=* ***pay-as-you-go IT resources via the internet.*** *•* ***IaaS*** *= raw VMs & networks • **PaaS** = code-ready runtimes • **SaaS** = finished apps •* ***Public / Private / Hybrid / Multi-Cloud*** *= where and how resources run • Always remember the* ***shared-responsibility line****—misconfigurations are still your problem.*

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit

Week 2: Day 1